4	**A**	The question asks for the HORIZONTAL forces only. The horizontal force the propels the man forward is the frictional force of pool’s floor on his feet. The horizontal resistive force on his motion comes from the drag due to water. Using Newton's second law along the horizontal direction: Net force on man = (mass of man)(acceleration of man) i.e. $f - f_D = ma$

5	A	Magnitude of the change in momentum Δp = area under graph = $\frac{1}{2}(150)(40 \times 10^{-3})$ = 3 N This